Sat. Sep 5th, 2026 **9:48:05 AM**

distherapy

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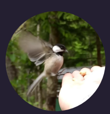
UNCATEGORIZED

Zero-Kill, Non-Agricultural System



By sixie6e

JUN 22, 2026

More_things

Using whatever is available, man-made or otherwise, we need to be able to exist without eating animals or killing them through agriculture. I'm tired of the primitive savages claiming stupid things such as cattle will *both* overpopulate *and* go extinct if they don't force-breed them and kill them. I'm tired of the back and forth articles and memes, the animal exploitation and brutality continue in the meantime. 40% of food goes to animals that humans use for food when we could skip that step entirely. Even if science or conscience proves they should have compassion, and that what they do is unnecessary, they will usually find excuse to continue their degenerate ways. Striving to achieve a truly zero-kill, zero-agriculture existence is the only way for humans to have any chance at ecological re-balancing. Christians think their zombie is coming back yet again to make everything okay. They don't care about the planet. In fact, many think that by rushing its destruction, their zombie will come back faster. Nationalists think the entire planet is theirs(even other countries), and they will scorch the Earth before letting anyone else play with their toy. Everything on the planet is there for them to do with what they like. I strive for minimal ecological impact regardless of their conditioning that is a blend of corporate and medieval mentalities.

Traditional agriculture—even strictly vegan farming—inherently relies on tilling land, which disrupts ecosystems, destroys topsoil, and causes high collateral mortality for small animals, birds, and insects. To eliminate both animal exploitation and the collateral damage of crop farming, we have to bypass the field entirely. By leveraging current technology, we can construct a completely non-agricultural, closed-loop nutritional framework. Instead of using plants to capture solar energy via photosynthesis, this approach relies on **chemosynthesis** via hydrogen-oxidizing bacteria (*Cupriavidus necator* or similar non-modified wild strains). Because it grows vertically in closed steel containers, its land footprint is practically zero—requiring only the space needed for solar panels or wind turbines to power the water-splitting electrolyzers.

While gas-fermented microbes provide the baseline bulk protein and fats, humans need specific functional macros to maintain long-term physical health without monotonous survival rations. Precision fermentation uses programmed micro-organisms (primarily specialized yeasts like *Pichia pastoris*) to brew bio-identical target nutrients without fields or animals. Structured fats can be brewed via oleaginous (fat-producing) yeast strains, yielding clean triglycerides, omega-3 fatty acids (DHA/EPA), and functional fats without harvesting fish or pressing oilseeds. Microalgae grown in dark, enclosed heterotrophic bioreactors can feed on basic acetic acid (which can be synthesized synthetically from CO2) to produce clean starches and functional dietary fibers for gut health.

To completely disconnect from soil-grown foods, the micronutrient architecture must be entirely elemental or synthetic.

Daily Framework

A complete nutrient delivery system designed to completely bypass the topsoil looks like this:

```
[Clean Energy + CO2 + Water]
        │
        ├──► Gas Fermentation  ──► Baseline Protein & B-Vitamins (Solein)
        ├──► Oleaginous Yeast  ──► Pure Essential Lipids & Omega-3s
        └──► Mix               ──► Geological Minerals & Synthetic Vitamins
        │
        ▼
[Total Nutritional Sustenance]
```

This model is a direct pipeline converting energy and abundant atmospheric molecules straight into human cells, bypassing the ecological extraction of traditional farming entirely. To sustain a human entirely on a non-agricultural, closed-loop bioreactor system, we have to look at the human body as a thermodynamic engine. To provide a standard diet of 2,000 kcal per day (730,000 kcal per year) using hydrogen-oxidizing bacteria (*Cupriavidus necator*) and precision-fermented yeasts, calculate the exact elemental mass and electricity required to drive the chemical synthesis.

Requirements

The entire system relies on indirect photosynthesis. Instead of plants using sunlight to split water, use electricity. The primary bottleneck is the efficiency of the bacteria in converting hydrogen gas into cellular energy, combined with the efficiency of water electrolysis. To yield 2,000 kcal of edible biomass per day, the bioreactors require approximately 15 to 18kWh of electrical energy per day, ~5,475 to 6,570kWh per person, per year. For perspective, this is roughly half the annual electricity consumption of an average American household, easily supplied by a modest 4 kW solar panel.

To build the macronutrients (proteins, lipids, carbs) for one person for a year, the bioreactors must be continuously fed pure, raw elements. Based on the average elemental composition of human biomass equivalents, here is the raw material manifest required per year:

Carbon Dioxide

Carbon is the backbone of all proteins, fats, and carbohydrates. The bacteria extract carbon directly from gaseous CO_2. 800 grams daily. *Source:* This can be captured directly from the atmosphere via Direct Air Capture (DAC) technology or reclaimed from your own exhaled breath in a sealed living environment.

Water

1.2kg daily. Water serves a dual purpose: it is the source of hydrogen (H_2) via electrolysis to feed the bacteria, and it forms the fluid medium of the bioreactor. *Note:* In a closed loop, much of this is recovered when your body metabolizes the food back into moisture and sweat, allowing for near-infinite recycling.

Fixed Nitrogen ($N_2 \rightarrow NH_3$)

Protein is rich in nitrogen. Unlike plants, which require vast fields and chemical fertilizers that ruin topsoil, these bioreactors are fed 35 grams of clean, aqueous ammonia or nitrates daily. *Source:* Synthesized from air via a miniature, cleanly powered Haber-Bosch system or reclaimed from processed metabolic waste.

Minerals & Micro-Nutrients

To keep both the microbes synthesizing correctly and cellular biology functioning, inorganic minerals must be added to the aqueous solution.

Annual Summary

Input Component	Mass Required (Per Year)	Primary Role
Electricity	5,500−6,500 kWh	Powering electrolysis & bioreactors
Water (H_2O)	438 kg	Splitting into H_2 fuel
Carbon Dioxide (CO_2)	292 kg	Building block for all macros
Nitrogen (N)	12.8 kg	Amino acid / Protein creation
Geological Minerals	~4 kg	Electrolytes, bone density, cell health

By organizing this setup, a single human can be sustained indefinitely in a space no larger than a small utility closet. The system outputs zero agricultural runoff, requires zero tilling of earth, and results in a completely calculated, mathematical zero-kill footprint. To transition this concept into a physical, operational reality within a small footprint:

1. A 4 kW to 6 kW photovoltaic array. Going slightly above the baseline 4 kW ensures adequate power generation during seasonal shifts or consecutive overcast days.

2. A 15 kWh to 20 kWh Lithium Iron Phosphate ($LiFePO_4$) battery bank. The bioreactors and gas delivery loops must run 24/7; you cannot rely solely on daytime generation.

3. A heavy-duty, off-grid hybrid inverter (48V) capable of managing sustained loads from pumps, heating elements, and the electrolyzer.

4. A Proton Exchange Membrane electrolyzer rated to produce enough hydrogen gas to sustain the *Cupriavidus necator* baseline. Look for an industrial or laboratory-grade unit capable of continuous duty cycles.

5. CO_2 System: *Open:* High-pressure food-grade CO_2 cylinders with dual-stage regulators. *Closed loop:* A miniature Direct Air Capture unit utilizing solid sorbent technology, or a closed-loop respiratory scrubbing integration that captures and purifies your own exhaled breath.

6. Specialized high-pressure storage tanks for hydrogen and oxygen. They are highly explosive when mixed, so you will need ATEX-rated explosion-proof fittings, optical gas sensors, and automated emergency shut-off valves.

7. A specialized 10L to 20L continuous-stirred tank bioreactor or bubble column reactor. It *must* be rated for safe gas-mixing and feature high-efficiency mass transfer impellers to dissolve the gases into the liquid medium.

8. Two distinct 5L to 10L stirred-tank bioreactors featuring precise temperature, pH, and dissolved oxygen control loops. One is dedicated to the oleaginous yeast (lipids) and the other to *Pichia pastoris* or heterotrophic microalgae.

9. A centralized controller such as a Raspberry Pi or Arduino running open-source automation software.

10. Laboratory or Tubular Bowl Centrifuge for harvesting the biomass from the liquid growth medium, separating the dense microbial cake from the supernatant.

11. A high-pressure homogenizer or an ultrasonic cell disruptor to break open the yeast and bacterial cell walls, releasing the functional proteins and lipids.

12. A small-scale vacuum dryer or spray dryer to turn the protein paste into a shelf-stable, dry powder. For lipids, a small solvent-free separation or mechanical oil-separation rig is needed to isolate pure triglycerides and omega-3s.

13. A large capacity chamber to sterilize growth media, bioreactor vessels, and tubing. Total sterility is the only barrier protecting your cultures from being overtaken by wild molds or bacteria.

14. Pure, verified biological strains of starter cultures acquired from a repository like the ATCC (American Type Culture Collection): *Cupriavidus necator* (for baseline protein/calories) *Pichia pastoris* (or specific oleaginous strains like *Yarrowia lipolytica* for structured lipids)

15. Chemicals: Aqueous Ammonia (NH_3)For the nitrogen source. Industrial lab-grade potassium phosphate, magnesium sulfate, calcium chloride, and trace elemental salts (iron, zinc, manganese, copper, cobalt). Food-grade sodium hydroxide and phosphoric acid to balance the bioreactor environments.

IMPORTANT: Building this setup requires integrating gas plumbing (Swagelok fittings are standard for hydrogen safety), electrical automation, and sterile technique. If you are aiming for a utility-closet scale, prioritizing modular industrial components over delicate laboratory glassware will ensure the durability needed for indefinite, continuous operation. Raw microbial slurry cannot be consumed directly in large quantities due to high nucleic acid content (which causes gout in humans) and the need to separate the target lipids and proteins. This system requires separate, dedicated environments for the gas fermentation and the precision fermentation loops, as they require vastly different gas-handling and sterile environments. *I am contacting some colleges and businesses to see if I can't find a way to make this happen, even if I am only involved as the 1 year trial subject. If I started ordering stuff in this list, I would probably have agents at my door.*

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Best Man for the Job »

By sixie6e

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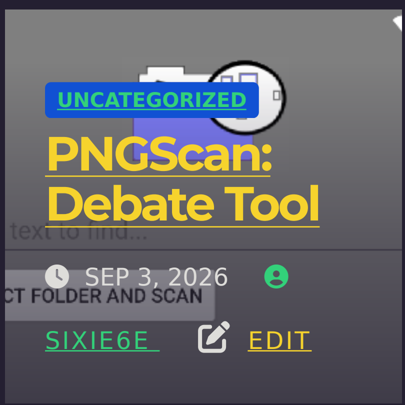

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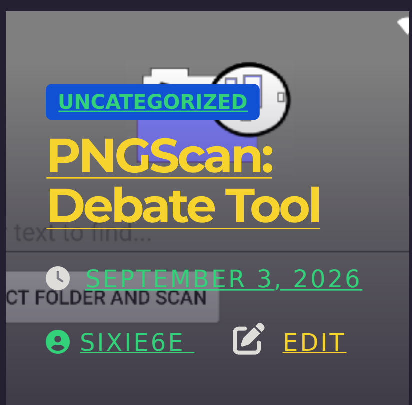

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